

LIPI — a programming language with Devanagari syntax

Hindi/Sanskrit keywords · compiled to a bytecode VM · written in 100% pure Rust · runs on desktop and in the browser (WebAssembly).

Most languages ask you to program in English. LIPI doesn't: keywords, error messages, and the standard library are Devanagari-native. Under the hood it's a real bytecode virtual machine — source compiles to opcodes, then executes.

Write a function, call it

```
विधि जोड़ो(अ, ब):  
    फल अ + ब  
  
बताओ जोड़ो(3, 4)
```

Output: 7

Batteries included — including things other stdlibs don't have

- Data & web: JSON, CSV, regex, HTTPS/TLS, sockets, threads, an HTTP server, SQL, ZIP.
- A mission-critical suite: units & dimensional analysis, linear algebra, quaternions, PID + Kalman control, geodesy, error-correcting codes, interval arithmetic.
- Arbitrary-precision integers — 50! exactly, no floating-point error:

```
आयात भारत.बड़ी  
बताओ महा_भाज्य("50")
```

Output: 30414093201713378043612608166064768844377641568960512000000000000

No ceiling: a C FFI

LIPI can call any C-ABI function in any native library — and hand a LIPI function back *as* a C callback. Here C's `qsort` sorts an array using a comparator written in LIPI:

```
आयात भारत.बाह्य  
पुस्तकालय है बाह्य_पुस्तकालय("msvcrt.dll")
```

```
# ... qsort called with a LIPI comparator as the C function pointer ...
```

Input: [5, 2, 8, 1, 9] → **Output:** [1, 2, 5, 8, 9]

Real HTTPS, so it can talk to modern APIs (including LLMs)

```
आयात भारत.सुरक्षित  
प है https_पाओ("https://api.github.com/zen", {"User-Agent": "LIPI"})  
बताओ प["स्थिति"]
```

Output: 200

Status: bytecode VM + WASM runtime, cross-platform FFI, 60+ passing example programs, an in-browser IDE, a formatter, linter, profiler, debugger, LSP, and a package manager. Open to contributors.